

SECURE CODING REVIEW REPORT

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Application Reviewed	PHP Web Application
Task	Secure Coding Review
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1. ABSTRACT

This report presents a secure coding review of a PHP-based web application. The objective of this assessment is to identify vulnerabilities, analyze their impact, and recommend remediation strategies based on OWASP security guidelines.

2. INTRODUCTION

Secure coding practices help protect applications from common cyber threats. This review focuses on identifying insecure coding patterns and improving application security.

3. METHODOLOGY

The review was conducted using manual code inspection and static analysis methods. Different attack vectors including SQL Injection, Cross-Site Scripting (XSS), and File Inclusion vulnerabilities were analyzed.

4. TOOLS USED

- PHP
- Grep Commands
- Manual Code Review
- Static Analysis
- OWASP Top 10 Reference

5. IDENTIFIED VULNERABILITIES

Vulnerability	Severity	Vulnerable Code	Recommended Fix	Impact
SQL Injection	High	SELECT * FROM users WHERE id = \$_GET[id]	Use prepared statements using PDO	Database compromise and data leakage
Cross-Site Scripting (XSS)	Medium	echo \$_GET['name']	Use htmlspecialchars()	Session hijacking and malicious scripts
File Inclusion	Medium	include(\$_GET['page'])	Whitelist allowed files	Possible remote code execution

6. SECURITY BEST PRACTICES

- Validate and sanitize all user inputs.
- Use parameterized queries and prepared statements.
- Enable HTTPS for secure communication.
- Apply least privilege access control.

- Keep frameworks and dependencies updated.
- Implement secure authentication mechanisms.
- Use proper logging and monitoring.

7. CONCLUSION

The secure coding review identified multiple vulnerabilities within the PHP web application. Recommended remediation techniques help improve application security and reduce cyber risks. Regular security assessments and secure development practices are strongly recommended.